Supplement S1: Mathematical Foundations

E_8 Exceptional Lie Algebra, G_2 Holonomy Manifolds, and K_7 Construction Brieuc de La Fournière

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E_8 Exceptional Lie Algebra, G_2 Holonomy Manifolds, and K_7 Construction

Mathematical background for the K_7 framework: the E_8 and G_2 data the dictionary uses, the topological status of K_7, and the algebraic reference form.

Lean: the arithmetic statements of this supplement are formalized in Arithmon/K7-Lean (the library rests on fourteen axioms beyond Mathlib's foundations, four literature imports and ten interval certificates, and uses no `sorry`; the 33 relations themselves depend only on the standard axioms; main §8, S3.7). Formalization establishes the arithmetic, not the physical reading. Lean names are given as they appear in the repository at commit `34ce857` (S3.7).

Abstract

This supplement presents the mathematical architecture underlying the K_7 framework. Part I gives the E_8 data the dictionary uses, with the arithmetic rewritings of earlier versions and their status. Part II introduces G_2 holonomy manifolds, including the characterization of the g_2 subalgebra as the kernel of the Lie derivative map. Part III discusses the topology of K_7: the pair $(b_2, b_3) = (21, 77)$ has no known construction. It does not appear among the catalogued compact G_2 manifolds, and the Joyce–Karigiannis $\mathbb{Z}_2{}^3$ route proposed in version 3.5 does not currently realise it, because its quotient cohomology is incompatible with the stated group action (§8.4). Part IV describes the algebraic reference form, whose scale is fixed by the imposed normalization $\det(g) = 65/32$ (main R2), and the status of Joyce's perturbation theorem for this setting: its smallness constant depends on geometric bounds that are not established here, and the existence of a compact torsion-free metric is open (main §9). Numerical (PINN) experiments are reported as such. The arithmetic statements are verified in Lean 4; an earlier module, `JoyceKarigiannisConstruction.lean`, verified arithmetic on an erroneous input and is withdrawn as evidence (§8.4).

Part 0: The Octonionic Foundation

0. Why This Framework Exists

The K_7 framework is motivated by a single algebraic fact:

The octonions $\mathbb{O}$ are the largest normed division algebra.

The chain of motivations, and where the framework's choices enter, is:

```
𝕆 (octonions, dim 8)
    |
    ▼
Im(𝕆) = ℝ⁷ (imaginary octonions)
    |
    ▼
G₂ = Aut(𝕆) (automorphism group, dim 14)
    |
    ▼
K₇ with G₂ holonomy (posited; seam metric certified in [A]; global existence open)
    |
    ▼
Topological invariants (b₂, b₃) = (21, 77)   ← input (main R1)
    |
    ▼
33 relations assigned to observables   ← dictionary (main §3), not derived
```

Status: the relations are assignments, written in the constants that the steps above supply; the dataset (95 entries, of which 33 relations are counted) is described in S3, and the arithmetic of each relation in S2. No geometric map from the invariants to the observables is known.

0.1 The Division Algebra Chain

Algebra	Dim	Physics Role	Stops?
$\mathbb{R}$	1	Classical mechanics	No
$\mathbb{C}$	2	Quantum mechanics	No
$\mathbb{H}$	4	Spin, Lorentz group	No
$\mathbb{O}$	**8**	**Exceptional structures**	**Yes**

The pattern terminates at $\mathbb{O}$. There is no 16-dimensional normed division algebra. The octonions are *the end of the line.*

0.2 G_2 as Octonionic Automorphisms

Definition: $G_2 = \{g \in GL(\mathbb{O}) : g(xy) = g(x)g(y) \text{ for all } x,y \in \mathbb{O}\}$

Property	Value	Role
$\dim(G_2)$	$14 = C(7,2) - C(7,1) = 21 - 7$	Q_{Koide} numerator
Action	Transitive on $S^6 \subset \text{Im}(\mathbb{O})$	Connects all directions
Embedding	$G_2 \subset SO(7)$	Preserves φ_0

0.3 Why $\dim(K_7) = 7$

The dimension 7 is a consequence of the octonionic structure, not an independent choice:

- $\text{Im}(\mathbb{O})$ has dimension 7
- G_2 acts naturally on $\mathbb{R}^7$
- G_2 occurs as a holonomy group only in dimension 7 (Berger, §6.3)

The choice made by the framework is G_2 holonomy; dimension 7 then follows.

0.4 The Fano Plane: Combinatorial Structure of $\text{Im}(\mathbb{O})$

The 7 imaginary octonion units form the **Fano plane** PG(2,2), the smallest projective plane:

- 7 points (imaginary units $e_1 \dots e_7$)
- 7 lines (multiplication triples $e_i \times e_j = \pm e_k$)
- 3 points per line

Combinatorial counts:

- Point-line incidences: $7 \times 3 = 21 = C(7,2) = b_2$
- Automorphism group: PSL(2,7) with $|\text{PSL}(2,7)| = 168$

Arithmetic identity:

$$(b_3 + \dim(G_2)) + b_3 = 91 + 77 = 168 = |\text{PSL}(2,7)| = \text{rank}(E_8) \times b_2$$

It is an identity among the chosen integers of the kind recorded in S2.7; no geometric structure is attached to it.

Part I: E_8 Exceptional Lie Algebra

1. Root System and Dynkin Diagram

1.1 Basic Data

Property	Value	Role
Dimension	$\dim(\mathrm{E}_8) = 248$	Gauge DOF
Rank	$\mathrm{rank}(\mathrm{E}_8) = 8$	Cartan subalgebra
Number of roots	$\lvert\Phi(\mathrm{E}_8)\rvert = 240$	E_8 kissing number
Root length	$\sqrt{2}$	α_s numerator
Coxeter number	$h = 30$	Icosahedron edges
Dual Coxeter number	$h^\vee = 30$	McKay correspondence

1.2 Root System Construction

E_8 root system in $\mathbb{R}^8$ has 240 roots:

Type I (112 roots): Permutations and sign changes of $(\pm 1, \pm 1, 0, 0, 0, 0, 0, 0)$

Type II (128 roots): Half-integer coordinates with even minus signs:

$$\frac{1}{2}(\pm 1, \pm 1, \pm 1, \pm 1, \pm 1, \pm 1, \pm 1, \pm 1)$$

Verification: $112 + 128 = 240$ roots, all length $\sqrt{2}$.

Lean: the root-system statements are proved:

- `D8_roots_card` = 112, `HalfInt_roots_card` = 128
- `E8_roots_card` = 240, `E8_roots_decomposition`
- `E8_inner_integral`, `E8_norm_sq_even`, `E8_sub_closed`
- `E8_basis_generates`: Every lattice vector is integer combination of simple roots (theorem)

1.3 Cartan Matrix

$$A_{E_8} = \begin{pmatrix} 2 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & -1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

Properties: $\det(A) = 1$ (unimodular), positive definite.

2. Weyl Group

2.1 Order and Factorization

$$|W(E_8)| = 696,729,600 = 2^{14} \times 3^5 \times 5^2 \times 7$$

2.2 Factorization in the Structural Constants

The Weyl group order can be written in the framework's structural constants:

$$|W(E_8)| = p_2^{\dim(G_2)} \times N_{gen}^{Weyl} \times Weyl^{p_2} \times \dim(K_7)$$

Factor	Exponent	Value	Origin
2^{14}	$\dim(G_2) = 14$	16384	$p_2^{\text{holonomy dim}}$
3^5	Weyl = 5	243	$N_{gen}^{\text{Weyl factor}}$
5^2	$p_2 = 2$	25	$\text{Weyl}^{\text{binary}}$
7^1	1	7	$\dim(K_7)$

This is an arithmetic rewriting of the prime factorization $2^{14} \cdot 3^5 \cdot 5^2 \cdot 7$, with each prime and exponent matched to a constant; it is the source of the constant Weyl = 5. Lean: `E8_weyl_order_factored`, `E8_weyl_factored` (the prime factorization).

2.3 Three Expressions for Weyl = 5

$$\frac{\dim(G_2) + 1}{N_{gen}} = \frac{b_2}{N_{gen}} - p_2 = \dim(G_2) - \text{rank}(E_8) - 1 = 5$$

Expression	Computation	Result
$(\dim(G_2) + 1)/N_{gen}$	$(14 + 1)/3$	5
$b_2/N_{gen} - p_2$	$21/3 - 2$	5
$\dim(G_2) - \text{rank}(E_8) - 1$	$14 - 8 - 1$	5

Earlier versions called this the "Weyl Triple Identity" and read the convergence as evidence that Weyl = 5 is "structurally constrained". Three expressions in small integers that evaluate to the same small integer are an arithmetic fact about the chosen integers; they fix no further quantity, and the reading is withdrawn (S2.1). Weyl enters $\det(g) = 65/32$ (main R2) and the factorization of §2.2. The "cosmological ratio $\sqrt{}$Weyl" of earlier versions belonged to the relation $\Omega_m = \Omega_{DE}/\sqrt{5}$, withdrawn in v3.6 (S2.6).

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3. Exceptional Chain

3.1 The Pattern

A pattern connects exceptional algebra dimensions to primes:

Algebra	n	dim(E_n)	Prime	Index
E_6	6	78	13	prime(6)
E_7	7	133	19	prime(8) = prime(rank(E_8))
E_8	8	248	31	prime(11) = prime(D_{bulk})

3.2 Three Factorizations

For n ∈ {6, 7, 8}:

$$\dim(E_n) = n \times prime(g(n))$$

where g(6) = 6, g(7) = rank(E_8) = 8, g(8) = D_{bulk} = 11 (= dim(M_4) + dim(K_7) = 4 + 7, the total bulk dimension of M-theory).

Check (Lean):

- E_6: 6 × 13 = 78 ✓
- E_7: 7 × 19 = 133 ✓
- E_8: 8 × 31 = 248 ✓

The index function g is chosen case by case (6, rank E_8, D_{bulk}), so the statement is a description of three factorizations, not a structural theorem. Lean: `all_exceptional_chain_relations_certified`. Earlier versions called this the "Exceptional Chain Theorem"; the name is withdrawn with the reading.

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4. $E_8 \times E_8$ Product Structure

4.1 Direct Sum

Property	Value
Dimension	496 = 248 × 2
Rank	16 = 8 × 2
Roots	480 = 240 × 2

4.2 τ Numerator Connection

The hierarchy parameter numerator:

$$\tau_{num} = 3472 = 7 \times 496 = \dim(K_7) \times \dim(E_8 \times E_8)$$

Lean: `tau_num_factorization`. τ is used only by Type II entries (S3).

4.3 The Binary Ratio $p_2 = 2$

p_2 is defined as $\dim(G_2)/\dim(K_7) = 14/7 = 2$ and is called the binary ratio. It is not a Pontryagin class [ERR, §6]. Earlier versions listed a "triple geometric origin" of p_2 (also 496/248, and the $\sqrt{2}$ of the E_8 root length); these are three different occurrences of the number 2, not three derivations of one constant, and the list is withdrawn.

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5. Exceptional Algebras from Octonions

The foundational role of octonions is established in Part 0. This section details the exceptional algebraic structures that emerge from $\mathbb{O}$.

5.1 Exceptional Jordan Algebra $J_3(\mathbb{O})$

Property	Value
$\dim(J_3(\mathbb{O}))$	$27 = 3^3$
$\dim(J_3(\mathbb{O})_0)$	26 (traceless)

E-series expression: 27 can be written in E-series dimensions:

$$\dim(J_3(\mathbb{O})) = \frac{\dim(E_8) - \dim(E_6) - \dim(SU_3)}{6} = \frac{248 - 78 - 8}{6} = \frac{162}{6} = 27$$

This is an arithmetic expression for 27 in E-series dimensions; $\dim J_3(\mathbb{O}) = 27$ is a Lie datum in its own right (main §3.1).

Lean: `j3o_e_series_certificate`.

5.2 F_4 Connection

F_4 is the automorphism group of $J_3(\mathbb{O})$:

$$\dim(F_4) = 52 = p_2^2 \times \alpha_{sum} = 4 \times 13$$

5.3 Exceptional Differences

Difference	Value	Framework expression
$\dim(\mathrm{E}_8) - \dim(\mathrm{J}_3(\mathbb{O}))$	$221 = 13 \times 17$	$\alpha_{\mathrm{sum}} \times (\dim \mathrm{G}_2 + \mathrm{N}_{\mathrm{gen}})$
$\dim(\mathrm{F}_4) - \dim(\mathrm{J}_3(\mathbb{O}))$	$25 = 5^2$	Weyl^2
$\dim(\mathrm{E}_6) - \dim(\mathrm{F}_4)$	26	$\dim(\mathrm{J}_3(\mathbb{O})_0)$

These are arithmetic rewritings (S2.7); they are not formalized as a group.

5.4 Structural Derivation of τ

The hierarchy parameter τ is defined as a ratio of framework constants:

$$\tau = \frac{\dim(E_8 \times E_8) \times b_2}{\dim(J_3(\mathbb{O})) \times H^*} = \frac{496 \times 21}{27 \times 99} = \frac{10416}{2673} = \frac{3472}{891}$$

Prime factorization:

- Numerator: $3472 = 2^4 \times 7 \times 31 = \dim(\mathrm{K}_7) \times \dim(\mathrm{E}_8 \times \mathrm{E}_8)$
- Denominator: $891 = 3^4 \times 11 = {\mathrm{N}_{\mathrm{gen}}}^4 \times \mathrm{D}_{\mathrm{bulk}}$

Alternative form: $\tau_{\mathrm{num}} = 7 \times 496 = \dim(\mathrm{K}_7) \times \dim(\mathrm{E}_8 \times \mathrm{E}_8) = 3472$

τ is used only by Type II entries (S3), which are not counted; earlier versions presented this definition as establishing τ as "a geometric constant rather than a free parameter", which a choice of ratio cannot do, and the sentence is withdrawn.

Lean: `tau_structural_certificate`.

Part II: G_2 Holonomy Manifolds

6. Definition and Properties

6.1 G_2 as Exceptional Holonomy

Property	Value	Role
$\dim(\mathrm{G}_2)$	14	$\mathrm{Q}_{\mathrm{Koide}}$ numerator
$\mathrm{rank}(\mathrm{G}_2)$	2	Lie rank
Definition	$\mathrm{Aut}(\mathbb{O})$	Octonion automorphisms

Lean: the G_2 cross-product statements are proved:

- `epsilon_antisymm`, `epsilon_diag`, `cross_apply` ✓
- `G2_cross_bilinear`, `G2_cross_antisymm`, `cross_self` ✓
- `G2_cross_norm` (Lagrange identity $\|u\times v\|^2 = \|u\|^2\|v\|^2 - \langle u,v\rangle^2$) ✓
- `reflect_preserves_lattice` (Weyl reflection) ✓
- `cross_is_octonion_structure` (case check over the 343 index triples) ✓
- `G2_equiv_characterizations` ✓ (by `rfl`: the two predicates are defined to coincide, so this is not a theorem about two characterizations)

6.2 G_2 as Kernel of the Lie Derivative

The G_2 subalgebra of so(7) admits a precise characterization as the stabilizer of the associative 3-form φ_0. For any antisymmetric matrix A in so(7), the Lie derivative of φ_0 is:

$$L_A(\varphi_0)_{ijk} = A_{ia}\varphi_{ajk} + A_{ja}\varphi_{iak} + A_{ka}\varphi_{ija}$$

The g_2 subalgebra consists of all A for which $L_A(\varphi_0) = 0$:

$$\mathfrak{g}_2 = \ker(L) = \{A \in \mathfrak{so}(7) : L_A(\varphi_0) = 0\}$$

This yields the decomposition so(7) = g_2 + V_7, where $\dim(g_2) = 14$ and $\dim(V_7) = 7$. The complement V_7 carries the standard 7-dimensional representation of G_2.

In practice, the kernel is computed via singular value decomposition (SVD) of the linear map L: so(7) –> $\Lambda^3(\mathbb{R}^7)$. The 14 singular vectors with eigenvalue zero span g_2; the 7 singular vectors with nonzero eigenvalue span V_7.

Note: A heuristic construction based on Fano-plane indices does not produce correct g_2 generators (each such generator is approximately 67% in g_2 and 33% in V_7). The kernel-based construction is the correct definition and must be used in all numerical computations involving g_2/V_7 decomposition.

6.3 Holonomy Classification (Berger)

Dimension	Holonomy	Geometry
7	**G_2**	**Exceptional**
8	Spin(7)	Exceptional

6.4 Torsion: Definition and Framework Interpretation

Mathematical definition: Torsion measures failure of G_2 structure to be parallel:

$$T = \nabla\phi \neq 0$$

For a G_2 structure φ, the intrinsic torsion decomposes into four irreducible G_2-modules:

$$T \in W_1 \oplus W_7 \oplus W_{14} \oplus W_{27}$$

Class	Dimension	Characterization
W_1	1	Scalar: $d\varphi = \tau_0 \star\varphi$
W_7	7	Vector: $d\varphi = 3\tau_1 \wedge \varphi$
W_{14}	14	Co-closed part of $d\star\varphi$
W_{27}	27	Traceless symmetric

Total dimension: $1 + 7 + 14 + 27 = 49 = 7^2 = \dim(K_7)^2$

The torsion-free condition requires all four classes to vanish simultaneously, a highly constrained state with 49 conditions.

Torsion-free condition:

$$\nabla\phi = 0 \Leftrightarrow d\phi = 0 \text{ and } d * \phi = 0$$

What the framework uses. None of the 33 relations uses a torsion value. The integer ${\kappa_T}^{-1} = 61 = b_3 - \dim(G_2) - p_2$ enters three relations (α^{-1}, Y_p, Ω_m/Ω_Λ); earlier versions called $\kappa_T = 1/61$ a "topological capacity for torsion" bounding deviations from the reference form, and said that physical interactions "emerge from the geometry of K_7". No computation connects 61 to torsion, and no mechanism is supplied for the second statement; both readings are withdrawn (main R3).

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7. Topological Invariants

7.1 Combinations Used by the Dictionary

Constant	Formula	Value	Status
det(g)	$p_2 + 1/(b_2 + \dim(G_2) - N_{gen})$	65/32	imposed normalization (main R2)
κ_T	$1/(b_3 - \dim(G_2) - p_2)$	1/61	integer combination (main R3)
$\sin^2\theta_W$	$b_2/(b_3 + \dim(G_2))$	3/13	relation A01 (S2.2)

det(g) and ${\kappa_T}^{-1}$ each have a second expression in the constants that agrees with the one above on the line $b_2 = 21$ and differs elsewhere (main R2, R3; S2.1).

7.2 The 61 Decomposition

$$\kappa_T^{-1} = 61 = \dim(F_4) + N_{gen}^2 = 52 + 9$$

Alternative:

$$61 = p_2 \times \text{Weyl} \times (\text{Weyl} + 1) + 1 = 2 \times 5 \times 6 + 1$$

Further expressions for 61 are arithmetic facts of the kind recorded in S2.7; they do not give 61 a physical meaning (main R3). Lean: `kappa_T_from_F4_generations`.

7.3 Spectral Geometry

The Laplace-Beltrami operator on K_7 admits a discrete spectrum with eigenvalues $0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \ldots$ Its first non-zero eigenvalue is λ_1 (the spectral gap).

A ratio of the constants:

$$\frac{\dim(G_2)}{H^*} = \frac{14}{b_2 + b_3 + 1} = \frac{14}{99} = 0.1414...$$

This is not the spectral gap, and it is not a bound on it; it is listed for comparison.

Analytical spectral gap: If K_7 were a twisted connected sum with a building block of $b_2(M_1) = 11$ (a decomposition that is not available, §8.3), the first eigenvalue of the scalar Laplacian admits a closed-form expression from the 1D Sturm-Liouville reduction on the TCS neck:

$$\boxed{\lambda_1 = \frac{\pi^2}{L^2 \cdot g_{ss}} = \frac{6\pi^2}{25 \cdot (b_2(M_1) + \text{rank}(E_8))} = \frac{6\pi^2}{475} = 0.12467...}$$

where L = 5 is the effective neck domain length (including ACyl tails), $g_{ss} = 19/6$ is the seam metric component of [A], written as $(b_2(M_1) + \text{rank}(E_8))/(3 \cdot \text{rank}(G_2))$, and $b_2(M_1) = 11$ is the Picard rank assigned to the first building block of the conjectured decomposition. A numerical computation on the seam metric (Richardson extrapolation, $N = 800 \to 1600$) gives $\lambda_1 = 0.12461$, 0.05% from the closed form.

This is a closed form for the first eigenvalue of the one-dimensional seam reduction, conditional on the identification above (see the status paragraph below).

Comparison with topological ratios:

Expression	Value	Deviation from λ_1	Status
$6\pi^2/475$ (closed form)	0.12467	0.00% (definition)	conditional on $b_2(M_1) = 11$
NK Richardson	0.12461	0.05%	Numerical verification
14/99 (ratio of constants)	0.14141	+13.4%	not a bound
13/99	0.13131	+5.3%	ratio of constants

Remark: The exact value $\lambda_1 = 6\pi^2/475$ involves π (from the Sturm-Liouville eigenvalue structure), not a rational number. The product $\lambda_1 \times H^* = 6\pi^2 \times 99/475 = 12.342$ (see the regularity below).

Lean: the arithmetic of the closed form is in the module `Spectral.PhysicalSpectralGap`; the analytic statements it relies on enter through the literature axioms of main §8. Earlier versions also described the near-identity $\dim(G_2)/\sqrt{2} \approx \pi^2$ (0.30%) as "resolved" by this formula, and a Lean module `Spectral.SelbergBridge` as connecting the spectral gap to a Dirichlet polynomial through the Selberg trace formula; neither bears on the framework, and both statements are withdrawn here.

Status of §7.3. The closed form assumes a TCS decomposition with $b_2(M_1) = 11$, whose building blocks are not identified (§8.3), and the numerical value is a spectrum of the approximate seam metric, not of a torsion-free metric on a compact K_7 (main §9). The regularity $\lambda_1 \cdot H^* \approx 12.34$ across the catalogued G_2 manifolds is an empirical observation of a scan (`canonical/scripts/construction_classification.py`), not a theorem. No relation of the framework uses λ_1.

7.4 Continued Fraction Structure

The ratio $\dim(G_2)/H^*$ has the continued fraction

$$\frac{14}{99} = [0; 7, 14] = \cfrac{1}{7 + \cfrac{1}{14}}$$

whose partial quotients are $7 = \dim(K_7)$ and $14 = \dim(G_2)$. This is a property of the ratio 14/99, not of the spectral gap. It is an arithmetic fact; no physical consequence is claimed.

7.5 Pell Equation Structure

H^* and $\dim(G_2)$ satisfy a Pell equation:

$$\boxed{H^{*2} - 50 \times \dim(G_2)^2 = 1}$$

Explicitly:

$$99^2 - 50 \times 14^2 = 9801 - 9800 = 1$$

where $50 = \dim(K_7)^2 + 1 = 49 + 1$.

Fundamental solution: $(99, 14)$ is the fundamental solution of $x^2 - 50y^2 = 1$. It is the square of $\varepsilon = 7 + \sqrt{50}$, which solves $x^2 - 50y^2 = -1$:

$$\varepsilon^2 = (7 + \sqrt{50})^2 = 99 + 14\sqrt{50}$$

Continued fraction: $\sqrt{50} = [7; \overline{14}]$, with period 1, which is why the same two integers appear in §7.4.

Status: arithmetic identity, verified in Lean; no physical consequence is claimed. Earlier versions combined it with a "selection principle" $\kappa = \pi^2/14$ (a spectral normalization in the Lean module `Spectral.SelectionPrinciple`, unrelated to the selection problem of main §1) into a link with the spectral gap; that reading is withdrawn.

Part III: K_7 Manifold Construction

8. Twisted Connected Sum Framework

8.1 TCS Construction

The twisted connected sum (TCS) construction provides the primary method for constructing compact G_2 manifolds from asymptotically cylindrical building blocks.

Compact G_2 manifolds can be built by gluing two asymptotically cylindrical (ACyl) G_2 manifolds along their cylindrical ends, with the topology controlled by a twist diffeomorphism φ.

8.2 Asymptotically Cylindrical G_2 Manifolds

Definition: A complete Riemannian 7-manifold (M, g) with G_2 holonomy is asymptotically cylindrical (ACyl) if there exists a compact subset $K \subset M$ such that M K is diffeomorphic to $(T_0, \infty) \times N$ for some compact 6-manifold N.

8.3 Topological Status of (21, 77)

K_7 is posited with $(b_2, b_3) = (21, 77)$ (main R1). **No construction of a compact G_2 manifold with these Betti numbers is currently known.** The Newton–Kantorovich certificate of [A] concerns the neck (seam) metric; it does not certify the Betti numbers, and no global torsion-free G_2 structure has been constructed (main §9).

Catalogued constructions. The pair does not appear among the catalogued compact G_2 manifolds (Joyce's orbifold constructions; the twisted connected sums of Corti–Haskins–Nordström–Pacini). Orthogonal TCS is excluded by parity ($b_2 + b_3 = 98$ is even; CHNP Lemma 6.7). Earlier versions added that a building-block scan (2026-04-14) showed that $b_2 = 21 > 20 = \max \rho(\mathrm{K3})$ forces a semi-Fano building block. In a twisted connected sum, $b_2(M) = \mathrm{rk}\, K_+ + \mathrm{rk}\, K_- + \mathrm{rk}(N_+ \cap N_-)$, where $K\pm$ are the kernels of the restriction maps $H^2(Z\pm) \to H^2(\mathrm{K3})$ and $N\pm$ their images (CHNP); b_2 is not bounded by the Picard rank of the K3 fibre, and the argument does not hold as stated. It is withdrawn: which building blocks and gluings could give $b_2 = 21$ is open. Non-orthogonal TCS and extra-twisted connected sums remain open within TCS.

Withdrawn: a lattice sketch. Earlier versions described sublattices N_1, N_2 of the K3 lattice $U^3 \oplus E_8(-1)^2$, of ranks 11 and 10, with $N_1 \oplus N_2 \oplus \langle 2 \rangle$ of full rank, as "consistent with a TCS construction". $N_1 \oplus N_2 \oplus \langle 2 \rangle$ cannot be an orthogonal decomposition of the unimodular K3 lattice ($\langle 2 \rangle$ is not unimodular), only a sublattice of finite index; and two building-block lattices $N_1 \perp N_2$ describe an orthogonal gluing, which is the case excluded by parity above. The embedding conditions were never verified and no matching data was constructed. The sketch is withdrawn as support for a TCS realisation.

The Joyce–Karigiannis route proposed in version 3.5 as a realisation of (21, 77) does not currently realise it (§8.4).

Global properties assumed, not established:

- Compact 7-manifold (no boundary)
- Simply connected: $\pi_1(K_7) = \{1\}$, $b_1 = 0$
- If a torsion-free G_2 structure g* exists on K_7, then $\mathrm{Hol}(g^*) = G_2$ [20, Prop. 11.2.3] and $\mathrm{Ric}(g^*) = 0$. Its existence is open (main §9; [ERR, §7.5]).
- (Euler characteristic $\chi(K_7) = 0$ is automatic for a closed odd-dimensional manifold; §9.1.)

Arithmetic. $b_2 = 21 = C(7,2)$ and $b_3 = 77 = C(7,3) + 2b_2 = 35 + 42$ are arithmetic facts about the posited integers; they define no geometric decomposition (§9.3).

8.4 The Joyce–Karigiannis $\mathbb{Z}_2{}^3$ Route: Proposed, Not Currently a Realisation

The Joyce–Karigiannis framework [43] resolves orbifolds $(T^3 \times K3)/G$, with $G \subset \mathrm{Aut}(T^3) \times \mathrm{Aut}(K3)$ acting with A_1-type singularities, by gluing families of Eguchi–Hanson spaces, and yields compact torsion-free G_2 manifolds. For a resolution of this type the Betti numbers are

$$b_2(N) = b_2\big((T^3 \times K3)/G\big) + b_0(\text{fixed loci}), \qquad b_3(N) = b_3\big((T^3 \times K3)/G\big) + b_1(\text{fixed loci}).$$

What version 3.5 claimed. For $G = \mathbb{Z}_2{}^3$ with a mixed-parity action, a four-phase computer-assisted audit (a symplectic V4 screen on a CI(2,2,2) K3, an anti-symplectic obstruction for P^5-linear maps, abstract existence of the K3 lattice action via Mukai [44], Garbagnati–Sarti [45] and Nikulin [51], and a fixed-locus table with contributions $(b_0, b_1) = (21, 55)$) was combined with a quotient cohomology $(b_2, b_3) = (0, 22)$ to give $b_2 = 0 + 21 = 21$ and $b_3 = 22 + 55 = 77$, on a K3 of Picard rank 1.

Why the claim does not hold. An audit of 2026-09-16 (Arithmon/K7, `docs/openwave-candidate/K7_P2_JK_postfreeze_cohomology_audit_2026_09_16.md`, with the standard-library checker `check_k3_jk_quotient_cohomology.py`) shows that the quotient cohomology is incompatible with the stated action:

- G acts diagonally on $T^3 \times K3$, so the rational cohomology of the quotient is the G-invariant part of the cohomology of $T^3 \times K3$. The stated trace of G on $H^2(K3)$ gives character multiplicities (5, 5, 2, 2, 2, 2, 2, 2), so $\dim H^2(K3; \mathbb{Q})^G = 5$.
- With $W = H^1(T^3; \mathbb{Q})$ a sum of three characters and $H^1(K3) = H^3(K3) = 0$, the Künneth formula gives $b_2((T^3 \times K3)/G) = 5 + \#\{i < j : \chi_i = \chi_j\} \geq 5$ and $b_3((T^3 \times K3)/G) \leq 16$. The checker enumerates all $8^3 = 512$ ordered torus-character triples and confirms both bounds; the product action compatible with the G_2 form gives $(b_1, b_2, b_3) = (1, 6, 16)$.
- The pair (0, 22) is therefore impossible for this action. Keeping the stated fixed-locus contributions (21, 55) and the same resolution formula would give $b_2 \geq 26$ and $b_3 \leq 71$, not (21, 77).
- The stated involution profile also requires a K3 of Picard rank at least 17 (`check_k3_jk_character_gate.py`; `K7_P2_neutrino_JK_character_obstruction.md`), not the Picard-rank-1 K3 on which version 3.5 placed it.

Scope. This invalidates this route as stated. It does not show that every Joyce–Karigiannis construction, or every construction of a compact G_2 manifold with Betti numbers (21, 77), is impossible. An earlier

Lean module, `JoyceKarigiannisConstruction.lean` (K7-Lean commit `6acbf49`, since removed; master theorem `jk_z23_construction_realizes_gift_betti`) verified the arithmetic of the Betti formula from the erroneous quotient input; it is withdrawn as evidence for the route. Correcting the route would require a fully specified product action, its quotient Betti numbers computed directly, and an independent audit of the singular components and of the resolution formula.

The neck. The NK-certified seam metric of [A] and the closed-form neck ansatz of [D] operate at the neck level of a putative compact extension and do not depend on which global construction, if any, completes it. They are evidence about a neck, not about the existence of K_7.

Intermediate-curvature splitting (Chen–Hong 2026). Chen and Hong [48] prove that, for certain (n, m), a noncompact manifold with nonnegative m-intermediate curvature admitting a smooth proper map of nonzero degree to $M^{n-m} \times T^{m-1} \times \mathbb{R}$ splits isometrically; outside that range they construct non-split examples on $S^{n-m} \times T^{m-1} \times \mathbb{R}$. For the neck topology $K3 \times T^2 \times \mathbb{R}$ ($n = 7$, $m = 3$) the splitting theorem does not apply. Nothing follows about the neck: being outside the range of a rigidity theorem does not show that non-split metrics with this curvature condition exist on $K3 \times T^2 \times \mathbb{R}$ (the examples are on $S^4 \times T^2 \times \mathbb{R}$). Earlier versions concluded that the Chebyshev corrections of the seam metric [A] are "not obstructed by a forced splitting"; that conclusion is withdrawn.

---

9. Cohomological Structure (Conditional on Existence)

9.1 Betti Spectrum, Poincaré Duality and Euler Characteristic

If a compact, simply connected K_7 with holonomy G_2 and $(b_2, b_3) = (21, 77)$ exists, Poincaré duality fixes its full Betti spectrum:

k	0	1	2	3	4	5	6	7
$b_k(K_7)$	1	0	21	77	77	21	0	1

with $b_1 = 0$ from simple connectivity, b_2 and b_3 posited (R1), and b_4 to b_7 by duality. The Euler characteristic of any compact oriented odd-dimensional manifold vanishes: $\chi(K_7) = 0$. Lean: `euler_char_K7_is_zero`, `poincare_duality_K7`. The combination $H^* = b_2 + b_3 + 1 = 99$ is used by the dictionary.

A Mayer–Vietoris computation of (b_2, b_3) would require a decomposition of K_7, for example the two halves of a twisted connected sum; none is available (§8.3).

9.2 Numerical Consistency

The spectral analysis of the approximate seam metric ([B]) reports 21 near-zero eigenvalues of the Laplacian on 2-forms and 77 on 3-forms, separated from the rest of the spectrum. These multiplicities are not independent evidence for the posited Betti numbers: [B] constructs the 2-forms from the split $11 + 10$

withdrawn in §8.3, and the 3-forms from the split 35 + 21 + 21, whose block of 35 is the identification withdrawn in §9.3. Nor are they a certificate, since the metric is approximate and is not a metric on a compact K_7.

9.3 The Integer 42

The number 42 appears in several relations as a combination of framework constants:

Expression	Value
$p_2 \times N_{gen} \times \dim(K_7)$	$2 \times 3 \times 7 = 42$
$2 \times b_2$	$2 \times 21 = 42$
$b_3 - C(7,3)$	$77 - 35 = 42$

It is not the Euler characteristic ($\chi(K_7) = 0$). Lean: `magic_42_gift_form`; at commit `34ce857` 42 is `two_b2 := 2 * b2`, and the earlier name `chi_K7` is kept only as a deprecated alias (S3.7).

Withdrawn: a "local ⊕ global" splitting of H^3. Earlier versions wrote $H^3(K_7) = H^3_{local} \oplus H^3_{global}$ with dimensions 35 + 42, the 35 identified with $\Lambda^3(\mathbb{R}^7)$. The 35 is the rank of the bundle $\Lambda^3 T^*K_7$, a pointwise dimension; it does not provide a 35-dimensional subspace of the global cohomology $H^3(K_7)$. The identity 35 + 42 = 77 is arithmetic and does not justify the decomposition.

Part IV: Metric Structure and Verification

10. Structural Metric Invariants

10.1 Structural Metric Invariants and Normalizations

Two quantities that the dictionary uses are written as properties of a metric. Neither is computed from a metric: ${\kappa_T}^{-1}$ is an integer combination and det(g) is a normalization imposed on the metric.

Invariant	Formula	Value	Status
κ_T	$1/(b_3 - \dim(G_2) - p_2)$	1/61	integer combination (main R3)
det(g)	$p_2 + 1/(b_2 + \dim(G_2) - N_{gen})$	65/32	imposed normalization (main R2; §10.3)

10.2 The Combination $\kappa_T = 1/61$

Arithmetic:

$$\kappa_T = \frac{1}{b_3 - \dim(G_2) - p_2} = \frac{1}{77 - 14 - 2} = \frac{1}{61}$$

Earlier versions read 61 as "effective matter degrees of freedom" (77 fermion modes minus 14 gauge constraints minus a binary factor). Nothing in the framework identifies b_3 with fermion modes or $\dim(G_2)$ with gauge constraints in this subtraction; the reading is withdrawn.

Status: integer combination (main R3)

10.3 Metric Determinant det(g) = 65/32

The value 65/32 can be written in several ways in the framework constants; three are:

Expression 1:

$$\det(g) = \frac{\text{Weyl} \times (\text{rank}(E_8) + \text{Weyl})}{2^{\text{Weyl}}} = \frac{5 \times 13}{32} = \frac{65}{32}$$

Expression 2 (the one used, main App. B):

$$\det(g) = p_2 + \frac{1}{b_2 + \dim(G_2) - N_{\text{gen}}} = 2 + \frac{1}{21 + 14 - 3} = 2 + \frac{1}{32} = \frac{65}{32}$$

Expression 3:

$$\det(g) = \frac{H^* - b_2 - 13}{32} = \frac{99 - 21 - 13}{32} = \frac{65}{32}$$

Expressions 1 and 2 agree on the line $b_2 = 21$ and differ elsewhere (main R2); expression 3 uses only b_3. Several expressions of one rational number in a few integers are not a derivation from topology. The actual chain is: the metric optimization was constrained to det(g) = 65/32 (a normalization target chosen to be compatible with the known structural parameters $g_{ss} = 19/6$, ${g_T}^2 = 7/6$), and the above formulas were identified post-hoc. Any rational number with small numerator/denominator can be expressed as combinations of a few integers.

Numerical value: 65/32 = 2.03125 (exact rational)

Status: imposed normalization (main R2), exact by construction and not derived from topology. Six of the 33 relations use $\det_{\text{num}}$ or $\det_{\text{den}}$ and depend on this choice (main §4.4).

---

11. Reference Form, Numerics and Formalization

11.1 The Algebraic Reference Form

Scaling law. A positive 3-form φ determines its metric g_φ through

$$(\iota_x\varphi) \wedge (\iota_y\varphi) \wedge \varphi = 6\, g_\varphi(x, y)\, \text{vol}_{g_\varphi}.$$

Under $\varphi \mapsto c\varphi$ ($c > 0$) the left side scales by c^3, and if $g \mapsto \lambda g$ the volume form scales by $\lambda^{7/2}$; hence $c^3 = \lambda^{9/2}$, that is

$$g_{c\varphi} = c^{2/3}\, g_\varphi, \qquad \det g_{c\varphi} = c^{14/3}\, \det g_\varphi.$$

The reference form. Fix a coframe $(e^1, \ldots, e^7)$ and let φ_0 be the standard form in it (§12.1); in that coframe the matrix of g_{φ_0} is I_7. To impose $\det g = D = 65/32$ in this fixed coframe,

$$\boxed{\varphi_{\mathrm{ref}} = D^{3/14}\,\varphi_0, \qquad g_{\mathrm{ref}} = D^{1/7}\,g_{\varphi_0}, \qquad \det g_{\mathrm{ref}} = D.}$$

Correction. Earlier versions wrote $\varphi_{\mathrm{ref}} = c\ \varphi_0$ with $c = D^{1/14}$ and $g_{\mathrm{ref}} = c^2\ I_7$, using the law $g_{c\varphi} = c^2\ g_\varphi$. That law is wrong; with the correct law, $D^{1/14}\ \varphi_0$ induces $\det g = D^{1/3} \approx 1.26645$, not $65/32$. The metric written there, $D^{1/7}\ I_7$, is the right metric, but it is induced by $D^{3/14}\ \varphi_0$.

Which metric the coframe is orthonormal for. The coframe is orthonormal for g_{φ_0}. For g_{ref} it is not: the matrix of g_{ref} in it is $D^{1/7}\ I_7$. In a coframe orthonormal for g_{ref} itself, the matrix of g_{ref} is I_7 and the form reads φ_0 again; the normalization $\det g = 65/32$ is a statement relative to the fixed coframe, not an invariant of the G_2 structure. "I_7" reflects this frame choice, not global flatness. The NK-certified model of [A] is a conditional statement about a seam region; it does not construct a compact K_7 (main §9).

Property	Value	Status
det(g)	65/32	exact by imposed normalization, not fitted to experiment (§10.3)
φ_{ref} components	7 of 35 non-zero	standard form (§12.2)
Joyce smallness hypothesis	depends on geometric bounds not established here	not verified (§11.2)

11.2 Joyce Existence Theorem and Global Solutions

Clarification: The reference form $\varphi_{\mathrm{ref}} = D^{3/14}\ \varphi_0$ is the standard form written in the fixed coframe of §11.1 (orthonormal for g_{φ_0}, not for g_{ref}), not a globally constant form on K_7. On a compact G_2 manifold, the coframe 1-forms $\{e^i\}$ satisfy $de^i \neq 0$ in general, so "constant components" does not imply $d\varphi = 0$ globally.

Actual solution structure: The topology and geometry of K_7 impose a deformation:

$$\varphi = \varphi_{\mathrm{ref}} + \delta\varphi$$

The torsion-free condition ($d\varphi = 0$, $d^*\varphi = 0$) is a **global constraint**. Joyce's perturbation theorem gives a nearby torsion-free G_2 metric when the torsion of a closed G_2 structure is small compared with a constant that depends on the injectivity radius and curvature bounds of that structure ([20], Thm. 11.6.1). Earlier versions compared a PINN torsion value ($\|T\|_{\max} = 4.46 \times 10^{-4}$) with a fixed threshold $\varepsilon_0 = 0.1$ and reported a "224× safety margin". There is no universal threshold: the constant has not been computed for this structure, and the comparison is not an existence argument (main §9).

(Version 3.5 added here that the bound $\kappa_T = 1/61$ places K_7 within Joyce's perturbative regime. No argument connects the number 1/61 to Joyce's smallness hypothesis, and the sentence is withdrawn; [ERR, §7.6].)

11.3 Independent Numerical Validation (PINN)

A companion numerical program trains physics-informed neural networks (PINNs) towards G_2 metrics on a model of K_7; since no compact K_7 is constructed, the model is an atlas, not a manifold whose existence is established. The three-chart atlas (neck + two Calabi-Yau bulk regions) uses approximately 10^6 trainable parameters in float64 precision.

Phase 2:

Metric	Value	Significance
$\|T\|_{max}$	4.46×10^{-4}	small; not compared with a computed Joyce constant
$\|T\|_{mean}$	9.8×10^{-5}	small
det(g) error	$< 10^{-6}$	the training constrains det(g) to 65/32, so this measures the constraint, not a confirmation

G_2 holonomy training (Phase 3, versions v2-v13):

Over successive training protocols, the training scores evolved as follows:

Metric	Initial (v5)	Current best (v11)	Improvement
`g2_self`	3.86	3.25	−16%
V_7 projection score	0.51	0.014	-97%
det(g) at neck	4.69	2.031	locked at target
phi drift	13.4%	0%	controlled

The `g2_self` and V_7 projection scores are training diagnostics of the companion code, and their definition is not stated here. Earlier versions described `g2_score` as a normalized projection of the Riemann curvature onto the complement of g_2, equal to 0 for G_2 holonomy and about 3.5 for the flat metric. Both cannot hold: a flat metric has $R = 0$, so a normalized projection of R vanishes or is undefined. And curvature valued in g_2 would give $Hol \subseteq G_2$, not $Hol = G_2$. The table records how the scores evolved; it is not a measure of holonomy.

A critical bug in the g_2 basis construction was discovered and corrected between versions 9 and 10: the Fano-plane heuristic does not produce correct g_2 generators. The correct g_2 subalgebra is the kernel of the Lie derivative map (Section 6.2). This correction led to significant improvement in all subsequent versions.

Earlier versions reported that the normalization target $\det(g) = 65/32$ "passes 8/8 independent consistency tests" (permutation, bootstrap, a Bayesian posterior, a joint p-value). A target that the optimization is constrained to cannot be validated by the optimization's output; these tests are withdrawn. $\det(g) = 65/32$ is imposed (main R2, §10.3).

11.4 Lean 4 Formalization

Scope of verification: The Lean formalization (Arithmon/K7-Lean; fourteen axioms, no `sorry`) verifies:

1. Arithmetic identities and algebraic relations between the framework's constants, including the 33 relations (`relations_v36_certified`, standard axioms only; S3.7)
2. Numerical enclosures (the ten interval certificates)
3. G_2 linear algebra on $\mathbb{R}^7$: exterior algebra, Hodge star, $\psi = \star\varphi$ (`Geometry` module)
4. Spectral modules (§7.3), which no relation uses

The axioms (see main §8): four prediction-chain axioms import literature results (`cheeger_inequality`, `literature_package`, `spectral_upper_bound`, `neck_dominates`), and ten interval certificates import numerical enclosures (`det_g_at_half`, `K3_eigenvalue_0...3` and their `_bracketed` forms). They carry content that cannot be checked by `decide` alone because it involves real arithmetic. The v3.5 text named four "external data-package" axioms, three of which had already been eliminated [ERR, §7.1].

It does **not** formalize:

- Existence of K_7 as a smooth G_2 manifold
- Physical interpretation of topological invariants
- Uniqueness of the TCS construction

```
-- GIFT.Foundations.AnalyticalMetric

def phi0_indices : List (Fin 7 × Fin 7 × Fin 7) :=
  [(0,1,2), (0,3,4), (0,5,6), (1,3,5), (1,4,6), (2,3,6), (2,4,5)]

def phi0_signs : List Int := [1, 1, 1, 1, -1, -1, -1]

def scale_factor_power_14 : Rat := 65 / 32

theorem torsion_satisfies_joyce :
  torsion_norm_constant_form < joyce_threshold_num := by native_decide

theorem det_g_equals_target :
  scale_factor_power_14 = det_g_target := rfl
```

Both statements are true as written and say less than their names. `torsion_satisfies_joyce` compares two integers, one of them a fixed threshold (288) that is not Joyce's constant, which has not been computed for this structure (§11.2); it is not an existence argument. `det_g_equals_target` is a definitional equality of the rational 65/32 with the target (§12.5).

11.5 The Chain of Choices

The logical structure from algebra to the reference metric, and where the framework's choices enter:

```
Octonions (O)
     |
```

```
     ▼
G_2 = Aut(O), dim = 14
     |
     ▼
Standard form φ_0 (Harvey-Lawson 1982)
     |
     ▼
φ_ref = D^{3/14} φ_0, D = 65/32  ← framework constraint (imposed, R2)
     |
     ▼
Metric g_ref = D^{1/7} g_{φ_0}    (law g_{cφ} = c^{2/3} g_φ)
     |
     ▼
det(g) = 65/32               ← imposed normalization, not fitted (§10.3)
     |
     ▼
sin^2θ_W = 3/13, Q = 2/3, ...  ← assigned (main §3); not functionals of this metric
```

---

12. Analytical G_2 Metric Details

12.1 The Standard Form φ_0

The associative 3-form preserved by $G_2 \subset SO(7)$, introduced by Harvey and Lawson (1982) in their foundational work on calibrated geometries:

$$\varphi_0 = \sum_{(i,j,k)\in\mathcal{I}} \sigma_{ijk}\, e^{ijk}$$

where:

- $\mathcal{I} = \{(0,1,2), (0,3,4), (0,5,6), (1,3,5), (1,4,6), (2,3,6), (2,4,5)\}$
- $\sigma = (+1, +1, +1, +1, -1, -1, -1)$

12.2 Linear Index Representation

In the C(7,3) = 35 basis:

Index	Triple	Sign	Index	Triple	Sign
0	(0,1,2)	+1	23	(1,4,6)	-1
9	(0,3,4)	+1	27	(2,3,6)	-1
14	(0,5,6)	+1	28	(2,4,5)	-1

Index	Triple	Sign	Index	Triple	Sign
20	(1,3,5)	+1	–	–	–

All other 28 components are exactly 0.

12.3 Metric Derivation

The metric of a positive 3-form is defined intrinsically by the identity of §11.1. Equivalently, with

$$B_\varphi(x,y) = \tfrac{1}{6}\,(\iota_x\varphi)\wedge(\iota_y\varphi)\wedge\varphi \in \Lambda^7,$$

written in the fixed coframe as $B_{ij}\ e^{1\cdots7}$, one has $g_\varphi = (\det B)^{-1/9}\ B$. For the standard φ_0, $B = I_7$ and $g = I_7$, $\det g = 1$; for $c\varphi_0$, $B = c^3\ I_7$ and $g = c^{2/3}\ I_7$, recovering the scaling law.

Earlier versions reconstructed the metric as $g_{ij} = (1/6)\ \Sigma_{k,l}\ \varphi_{ikl}\ \varphi_{jkl}$. That contraction raises indices with a background Euclidean metric, so it is not a general reconstruction of g_φ: it agrees with g_φ only when φ is already standard for that background metric, and for $c\varphi_0$ it gives $c^2\ I_7$ instead of $c^{2/3}\ I_7$. This is the source of the error corrected in §11.1. Setting $c^{14/3} = 65/32$, i.e. $c = (65/32)^{3/14}$, yields the framework's reference metric.

12.4 Comparison: Fano Plane vs G_2 Form

Structure	7 Triples	Role
Fano lines	(0,1,3), (1,2,4), (2,3,5), (3,4,6), (4,5,0), (5,6,1), (6,0,2)	G_2 cross-product ε_{ijk}
G_2 form	(0,1,2), (0,3,4), (0,5,6), (1,3,5), (1,4,6), (2,3,6), (2,4,5)	Associative 3-form

Both have 7 terms because both are Fano-plane patterns: the associative calibration is $\varphi(x, y, z) = \langle x \times y, z\rangle$ on $\mathrm{Im}(\mathbb{O})$, and the two tables differ by a relabelling of the seven points. The training bug came from penalising the wrong labelling, not from a difference in kind between the two objects.

12.5 Verification Summary

Method	Result	Reference
Algebraic	$\varphi_{ref} = (65/32)^{3/14}\ \varphi_0$, $g_{ref} = (65/32)^{1/7}\ g_{\varphi_0}$	§11.1

Method	Result	Reference
Lean 4	`det_g_equals_target : rfl` (a definitional equality of the rational 65/32 with the target; it does not check the scaling law)	AnalyticalMetric.lean
PINN	small torsion, holonomy score improved; not a torsion-free metric	§11.3
Joyce theorem	applies under a smallness hypothesis whose constant is not computed here	[20]; main §9

None of these methods establishes a torsion-free metric on a compact K_7 (main §9).

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Author's Related Works

- **[A]** B. de La Fournière, "A Certified Torsion-Free G_2 Structure on a TCS Neck Model via Computer-Assisted Proof," Zenodo 10.5281/zenodo.18643069 (2026).
- **[B]** B. de La Fournière, "Spectral Geometry of an Explicit G_2 Metric: Laplacian Spectrum and Harmonic Forms," Zenodo 10.5281/zenodo.18920367 (2026).
- **[D]** B. de La Fournière, "An Explicit Closed-Form G_2 Ansatz on a K3-Coassociative Neck with Hyperkähler Rotation and Picard-Lefschetz Wirtinger Certificate," Zenodo 10.5281/zenodo.20039065 (2026).
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The K_7 Framework (formerly GIFT): Supplement S1 Mathematical Foundations: E_8 + G_2 + K_7

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